Software Test Strategy and Test Plan — Template

IEM Admission Management System

Code Alpha – Group No. (5) | Section A – 2024–2028 | Date: 04-10-2026

PART A — SOFTWARE TEST STRATEGY DOCUMENT

1. Introduction and Objectives

The purpose of testing the IEM Admission Management System is to verify and validate the complete admission process. Objectives: verify registration/login, personal and academic information, document upload/verification, application submission, merit calculation/list generation, seat allotment, payment after seat allotment, seat acceptance/decline and receipt generation; identify defects; and ensure security, reliability, usability and correct performance.

2. Scope

Testing covers User Registration, Login, Personal Information, Academic Information, Document Upload, Document Verification, Application Submission/Status, Merit Calculation, Category-wise Merit List, Seat Allotment, Payment Processing/Status, Seat Acceptance/Decline, Receipt Generation/Download, Notification Service and Admin Application Management.

3. Levels of Testing

3.1. Unit Testing

Individual modules/functions such as login(), saveDraft(), submit(), approve(remark), calculateMerit() and generatePDF() will be tested separately using valid, invalid and boundary inputs. Developers will perform unit testing.

3.2. Integration Testing

Bottom-up integration will be used. Modules will be combined gradually: Registration → Login → Application → Document Upload → Document Verification → Merit Calculation → Seat Allotment → Payment → Seat Acceptance → Receipt Generation. Stubs/drivers will be used where required.

3.3. System Testing

The fully integrated system will be tested against requirements using functional, performance, security, usability, regression and end-to-end testing.

3.4. Acceptance Testing

Alpha testing will be performed internally by the team. Beta/user acceptance testing will use representative students and admission/admin users. Acceptance requires correct major functions and no critical defects.

4. Testing Techniques Used for Test-Case Design

4.1. Black-box techniques

- Equivalence Class Partitioning: divide inputs into valid/invalid classes.
- Boundary Value Analysis: test values at and around limits.
- Decision Table Testing: test document verification, merit eligibility and seat availability combinations.
- State Transition Testing: Draft → Submitted → Under Review → Approved/Rejected → Merit Listed → Seat Allotted → Accepted/Declined.

4.2. White-box techniques

- Statement Coverage: execute each important statement.
- Branch/Decision Coverage: execute both outcomes of decisions.

- Basis Path/Path Coverage: cover independent paths in document verification and seat allotment.

5. Testing Types

- Functional: all admission functions.
- Regression: after changes to merit/document/seat modules.
- Performance/load: simultaneous login, upload, submission, merit checking and receipt download.
- Security: authentication, authorization, payment/status and document/data access.
- Usability: navigation and ease of completing admission tasks.

6. Debugging Approach

Cause elimination will be used: reproduce failure, locate module, analyze cause, fix code, rerun failed test, perform regression testing and record result.

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7. Entry and Exit Criteria

Testing Level	Entry Criteria	Exit Criteria
Unit Testing	Module developed	All planned unit tests pass; no critical defect
Integration Testing	Required modules unit-tested	Interfaces work; integration defects resolved
System Testing	Complete system integrated	Major functional/non-functional tests pass
Acceptance Testing	System testing complete	User/client accepts; no critical defect

8. Roles and Responsibilities

Role	Responsibility
Developer	Develop modules and fix defects
Test Engineer	Design and execute test cases
Test Lead	Prepare strategy, monitor testing and review results
Client / End-user Rep.	Validate functionality and perform acceptance testing

9. Test Tools and Environment

Tools: Visual Studio Code, web browser, PostgreSQL, Git/GitHub, Postman and browser developer tools. Testing: manual testing, API testing with Postman and automation where required. Environment: React.js frontend; Node.js/ Express.js backend; PostgreSQL database; AWS S3 file storage; JWT authentication.

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10. Risk-Based Considerations

- Incorrect merit calculation — extra testing of scores, categories and boundaries.
- Document verification failure — test valid, invalid, rejected and re-uploaded documents.
- Incorrect seat allotment — test different merit positions, categories and seat availability.
- Payment failure/status mismatch — test successful, failed, cancelled and pending payment before seat acceptance.

PART B — SOFTWARE TEST PLAN

1. Test Plan Identifier

Test Plan ID: IAMS-TP-01 | Project: IEM Admission Management System | Version: 1.0

2. Objective

Verify student application, document verification, merit calculation, merit list generation, seat allotment, payment after seat allotment, seat acceptance/decline and receipt generation from application submission to final confirmation.

3. Features to be Tested

- Student Registration
- Student Login
- Personal Information
- Academic Information
- Document Upload
- Document Verification
- Application Submission
- Merit Calculation
- Merit List Generation
- Seat Allotment
- Payment Processing/Status
- Seat Acceptance/Decline
- Receipt Generation/Download
- Admin Application Management
- Notifications

4. Features Not to be Tested (Out of Scope)

- Actual online banking/payment gateway processing and physical verification of original documents are outside this test plan. The system's payment initiation, success/failure/pending status handling, confirmation and receipt generation are within scope.

5. Business Rule(s) / Feature Under Test

A student must complete the application and submit required information/documents. Admin verifies documents. Rejected documents must be replaced. After verification, merit is calculated and a category-wise merit list is generated. Eligible candidates receive available seats based on merit and applicable category rules. After a seat is allotted, the student must complete the required payment successfully before accepting the seat. A successful payment allows seat acceptance.Successful seat acceptance generates the admission/seat confirmation receipt. Failed, cancelled or pending payment must not allow final seat acceptance.

6. Black-Box Test Design — Equivalence Partitioning & Boundary Value Analysis

Input under test: Entrance/Academic Score; assumed valid range for testing: 0–100.

Class Type	Description	Representative Value(s)
Valid	Within valid range	50
Valid Boundary	Minimum	0
Valid Boundary	Maximum	100
Invalid	Below minimum	-1
Invalid	Above maximum	101

TC ID	Description	Input (Entrance Score)	Expected Output	Actual Output	Pass/ Fail
TC-01	Normal valid score	50	Score accepted	—	—
TC-02	Minimum boundary	0	Score accepted	—	—
TC-03	Maximum boundary	100	Score accepted	—	—
TC-04	Below minimum	-1	Score rejected	—	—
TC-05	Above maximum	101	Score rejected	—	—

7. Black-Box Test Design — Decision Table Testing

Conditions: C1 = Documents verified?; C2 = Candidate eligible according to merit?; C3 = Seat available?; C4 = Payment successful?. Action: Assign Seat / Payment Required / No seat / Do not assign / Request verification / Reject-hold / Allow Acceptance / Generate Receipt.

Rule	C1	C2	C3	C4	Action
R1	Yes	Yes	Yes	No	Assign Seat; Payment Required
R2	Yes	Yes	Yes	Yes	Allow seat acceptanceNo seat / waiting
R3	Yes	Yes	No	—	No seat / waiting
R4	Yes	No	Yes	—	Do not assign
R5	No	Yes	Yes	—	Request verification
R6	No	Yes	No	—	Request verification
R7	No	No	Yes	—	Reject/hold
R8	No	No	No	—	Reject/hold

TC ID	Input Set 1 (Documents)	Input Set 2 (Merit)	Input Set 3 (Seat)	Payment	Expected Result
TC-06	Verified	Eligible	Available	Not Paid	Seat allotted; payment required
TC-07	Verified	Eligible	Available	Paid	Seat acceptance enabled
TC-08	Verified	Eligible	Not Available	—	No seat allotted
TC-09	Verified	Not Eligible	Available	—	No seat allotted
TC-10	Not Verified	Eligible	Available	—	Verification required
TC-11	Verified	Eligible	Available	Failed	Seat acceptance blocked; payment retry/status shown
TC-12	Verified	Eligible	Available	Pending	Seat acceptance blocked until payment success

8. White-Box Test Design — Basis Path Testing

Module under test: Document Verification.

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Node 1  : Start
Node 2  : Receive document
Node 3  : Check upload
Node 4  : not uploaded
Node 5  : mark pending
Node 6  : uploaded
Node 7  : check validity
Node 8  : valid
Node 9  : approve
Node 10 : invalid
Node 11 : reject
Node 12 : end
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Control flow graph: Nodes = {1 Start; 2 Receive document; 3 Check upload; 4 not uploaded; 5 mark pending; 6 uploaded; 7 check validity; 8 valid; 9 approve; 10 invalid; 11 reject; 12 end}; Edges: 1→2, 2→3, 3→4, 4→5, 4→6, 6→7, 7→8, 8→9, 8→10, 10→11, 5→12, 9→12, 11→12. Cyclomatic complexity V(G) = decision nodes + 1 = 2 + 1 = 3. Number of independent paths to cover = 3.

Path #	Path Description	Input Values	Expected Result
P1	Document not uploaded	No document	Status = Pending
P2	Uploaded and valid	Valid document	Document Approved
P3	Uploaded but invalid	Invalid document	Document Rejected

9. Integration Test Cases

Integration strategy: Bottom-up. Stubs/drivers used: where required.

TC ID	Integration Step	Description	Expected Result
INT-01	Registration + Login	Register and login	Student logs in
INT-02	Login + Application	Login and open application	Application opens
INT-03	Application + Documents	Upload documents	Documents linked
INT-04	Documents + Verification	Admin verifies	Status updated
INT-05	Verification + Merit	Verified application to merit	Merit calculated
INT-06	Merit + Seat	Eligible candidate processed	Seat allotted
INT-07	Seat + Payment	Student completes payment for allotted seat	Payment status = Successful
INT-08	Payment + Acceptance	Student accepts seat after successful payment	Status = Accepted
INT-09	Acceptance + Receipt	Accepted seat processed	Receipt generated
INT-10	Receipt + Download	Student downloads generated receipt	Receipt downloaded successfully

10. System Test Cases (End-to-End, Black-Box)

TC ID	Scenario	Steps	Expected Result
SYS-01	Registration	Register with details and submit	Account created
SYS-02	Login	Enter valid credentials	Dashboard displayed
SYS-03	Invalid Login	Enter incorrect credentials	Error displayed
SYS-04	Application Submission	Enter info, upload documents, submit	Application ID generated
SYS-05	Save Draft	Fill partially and save	Draft saved
SYS-06	Document Verification	Admin checks and approves	Approved status
SYS-07	Rejected Document	Admin rejects with remark	Student notified
SYS-08	Replace Document	Student uploads replacement	Replacement submitted
SYS-09	Merit Calculation	Admin calculates merit	Merit position generated
SYS-10	Merit List	Admin generates category-wise list	List generated
SYS-11	Seat Allotment	Eligible candidate processed	Seat allotted; payment option enabled
SYS-12	Payment — Success	Student selects payment and completes it successfully	Payment successful; seat acceptance enabled
SYS-13	Payment — Failure	Student payment fails	Payment failed; seat acceptance remains blocked
SYS-14	Payment — Pending/Cancelled	Payment remains pending or is cancelled	Seat acceptance remains blocked until successful payment
SYS-15	Accept Seat	Student accepts seat after successful payment	Status accepted
SYS-16	Decline Seat	Student declines seat	Status declined
SYS-17	Receipt Generation	Student accepts paid seat	Receipt generated
SYS-18	Download Receipt	Click download	Receipt downloaded

11. Test Deliverables

- Test plan document
- Test case documents
- Test execution/results log
- Defect reports
- Test summary report

12. Suspension and Resumption Criteria

Suspend: severity-1/blocker defect, application cannot launch, database unavailable, major functionality unavailable payment-status integration unavailable or unstable test environment. Resume: critical defect fixed, environment restored, required modules available and smoke testing passes.

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13. Approval

Name	Role	Signature	Date
Subho Pramanick	Test Lead		
Ranajit Chattaraj	Project Manager		
Aratreeka Pahari	Test Engineer		
Swapna Swadhikta Behera	Client/User Representative		